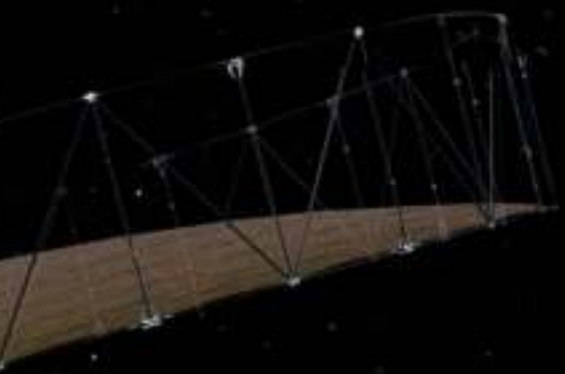
2,392
KG
Weight of NISAR satellite

743
KM
The Sun-synchronous orbit into which NISAR will be injected

12
METRES
(39 feet) Size of NISAR's unfurlable mesh reflector antenna

242
KM
NISAR's imaging swathe, which will allow it to cover the entire planet every 12 days



THE MISSION GOAL

To better understand...

- Earthquakes
- Volcanic eruptions
- Oil spills
- Floods
- Wildfires
- Landslides



HOW SAR WORKS

- Synthetic Aperture Radar (SAR) transmits microwave pulses and receives the echoes of the returned pulses
- Images every point on the ground in the path of its swathe many times
- Imagery is created from the reaction of an emitted pulse with physical structures like mountains, forests, and sea ice and conditions like soil moisture
- It can therefore see through clouds and darkness, and collect data in any weather
- The 12-metre antenna reflector, made of a gold-plated wire mesh, focuses the signals emitted and received

